6 **(a)** In 1919, Rutherford performed the first nuclear reaction induced in a laboratory in which a stationary nitrogen nucleus $^{14}_{7}N$ bombarded with an α-particle of a certain energy, transmutes to an oxygen nucleus $^{17}_{8}O$ and a proton.

Data:
mass of $^{14}_{7}N$ = 13.9993 u;
mass of $^{17}_{8}O$ = 16.9947 u;
mass of a proton = 1.0073 u;
mass of an α-particle = 4.0015 u.

(i) Write an equation for this nuclear reaction, showing the mass numbers and the atomic numbers of the particles involved.

[2]

(ii) Calculate the minimum kinetic energy of the alpha particle for the reaction to make this reaction occur.

kinetic energy = ……J [3]

(b) A radioactive isotope of thallium $^{207}_{81}Tl$ emits a β-particle and is thought to emit a gamma photon. The half-life of $^{207}_{81}Tl$ is 135 days.

(i) The radiation is allowed to pass though perpendicularly a vertical uniform magnetic field and the photographs of traces is obtained in a cloud chamber under certain conditions. Fig 6.1 show tracks produced by the beta-particles and gamma ray photons.

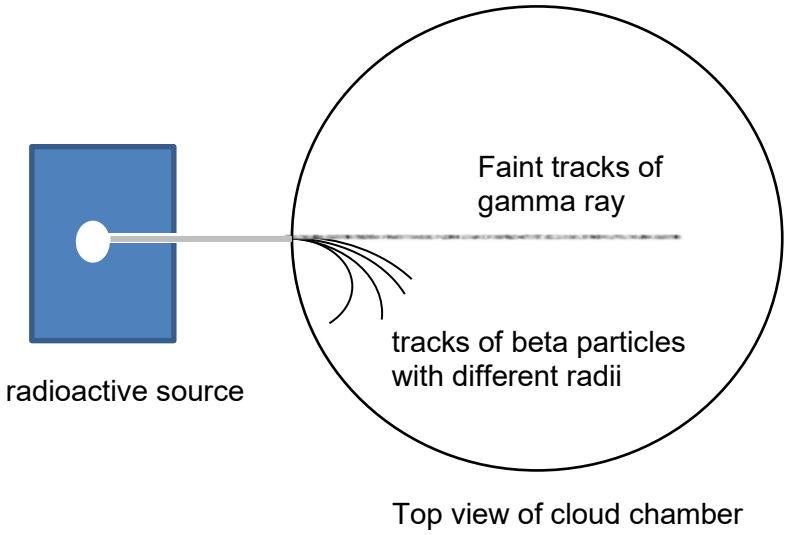

Fig. 6.1

Explain the features of the tracks.

……

……

……

……

……

…… [3]

(ii) An isotope of thallium $^{207}_{81}Tl$ emits a β-particle with an average energy of 2.4×10^{-13} J.

Calculate

1. the total energy available from 1 g of thallium-207,

total energy = …….J [2]

2. the initial rate at which the β-particles are emitted from 1 g of the freshly prepared isotope,

initial rate of emission = …..…….day^{-1} [2]

3. the initial power available from the beta particles emitted at the rate calculated in **b(ii)2**.

initial power = ……. W [1]

[Turn over